

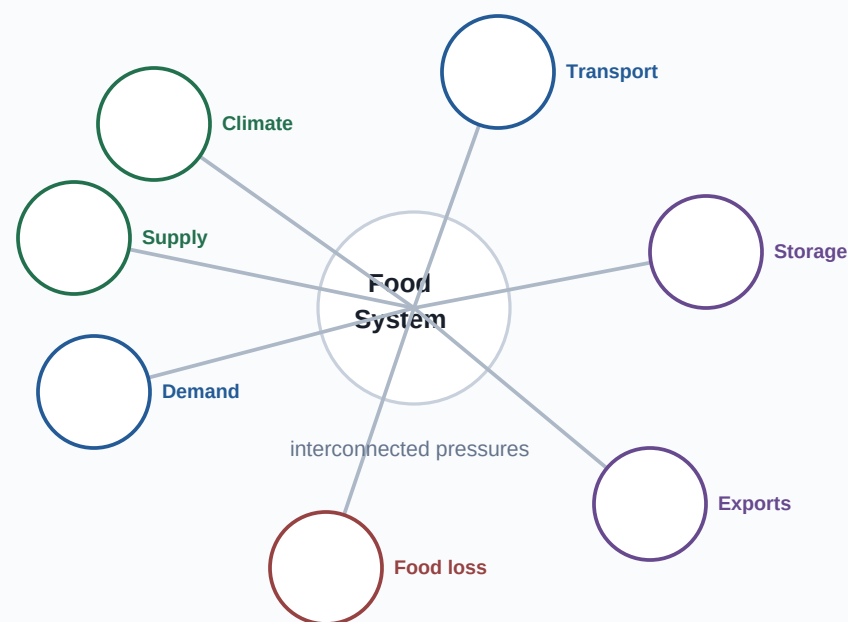
Systems Under Pressure

The Mathematics of Food Systems

"All models are wrong, but some are useful."

George Box

A spacious mathematical field notebook for sketching, modeling, revising, and explaining.



Rough work belongs here.

This journal is not a worksheet packet. It is the visible record of your group's evolving mathematical thinking.

Value uncertainty

Label assumptions, questions, disagreements, and places where a model stops making sense.

Revise visibly

Cross out, annotate, draw arrows, compare versions, and explain why your thinking changed.

FIELD NOTE STYLE

Messy is acceptable. Hidden thinking is not. The strongest journals show graph, algebra, context, units, limitations, and revision together.

Use the journal to make standards visible.

The rubric does not only score final answers. It looks for reasoning, interpretation, revision, limitations, precision, and flexible thinking across the investigation.

CORE CONTENT STANDARDS

1. Sequences and Series

discrete change,
arithmetic vs geometric
patterns, recursive vs
explicit forms, finite vs
infinite reasoning,
cumulative sums in
context

2. Exponentials and Logarithms

growth and decay, growth
factors, thresholds,
logarithmic reasoning,
realistic interpretation
and limitations

3. Polynomials

degree, leading
coefficient, end
behavior, intercepts,
multiplicity, extrema,
polynomial forms, and
realistic domain
restrictions

SKILL GRADES

4. Mathematical Communication

clear explanations,
graph-algebra-context
connections,
assumptions,
annotations, tables, and
contextual
interpretation

5. Comparing Models, Assumptions

model comparison,
realism, hidden
variables, tradeoffs,
assumptions,
unreasonable
predictions, and
meaningful revision

6. Algebraic Precision and Flexible Thinking

accurate methods,
factoring, logarithms,
substitution, graph
interpretation, rational
numbers, units, and
reasonableness checks

CLASS 1

Entering the System

What relationships are we beginning to notice?

Use this section as a field notebook. Let rough work, uncertainty, recalculation, and revision stay visible.

EXPLORATORY FIELD NOTES

Open, uncertain, observational.

Sketch variables, links, pressures, unknowns, and possible feedback

What changed in your

Original idea:

Revised idea:

Reason for change:

Variable table and measurement notes

Variable | Units | Why it matters | What is hard to measure

EXPLAIN WHY

Do not only write the model.
Explain why this model fits
and where it fails.

open canvas

Sketch before technology

Predict the shape of one relationship before using Desmos, Sheets, or another tool.

REFLECTION PROMPT

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

REVISION MARKER

Original idea:

Revised idea:

Reason for change:

Vlog notes

Patterns noticed, uncertainties, assumptions, important variables.

EXPLAIN WHY

Do not only write the model.
Explain why this model fits
and where it fails.

open canvas

CLASS 2

Modeling Change

How is the system changing over time?

Use this section as a field notebook. Let rough work, uncertainty, recalculation, and revision stay visible.

PATTERN STUDIO

Structured repeated change.

Discrete vs continuous comparison

Why is a sequence appropriate? What would change in a continuous model?

student thinking space

REFLECTION PROMPT

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

REVISION MARKER

Original idea:

Revised idea:

Reason for change:

Use evidence. Explain what each term represents in context.

Do not only write the model.
Explain why this model fits
and where it fails.

Recursive and explicit formulas

Write both formulas. Annotate every variable and parameter.

REFLECTION PROMPT

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

REVISION MARKER

Original idea:

Revised idea:

Reason for change:

Model versions

Compare 0-5, 0-10, and 0-20. What changes in realism and reliability?

REFLECTION PROMPT

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

REVISION MARKER

Original idea:

Revised idea:

Reason for change:

Changing assumptions creates different mathematical realities. Compare at least three related model versions.

[illegible]

CLASS 3

Acceleration, Decay, and T

When does repeated change become accelerated or compounding?

DYNAMIC MODELING

Rates reshape thresholds.

Use this section as a field notebook. Let rough work, uncertainty, recalculation, and revision stay visible.

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Reason for change:

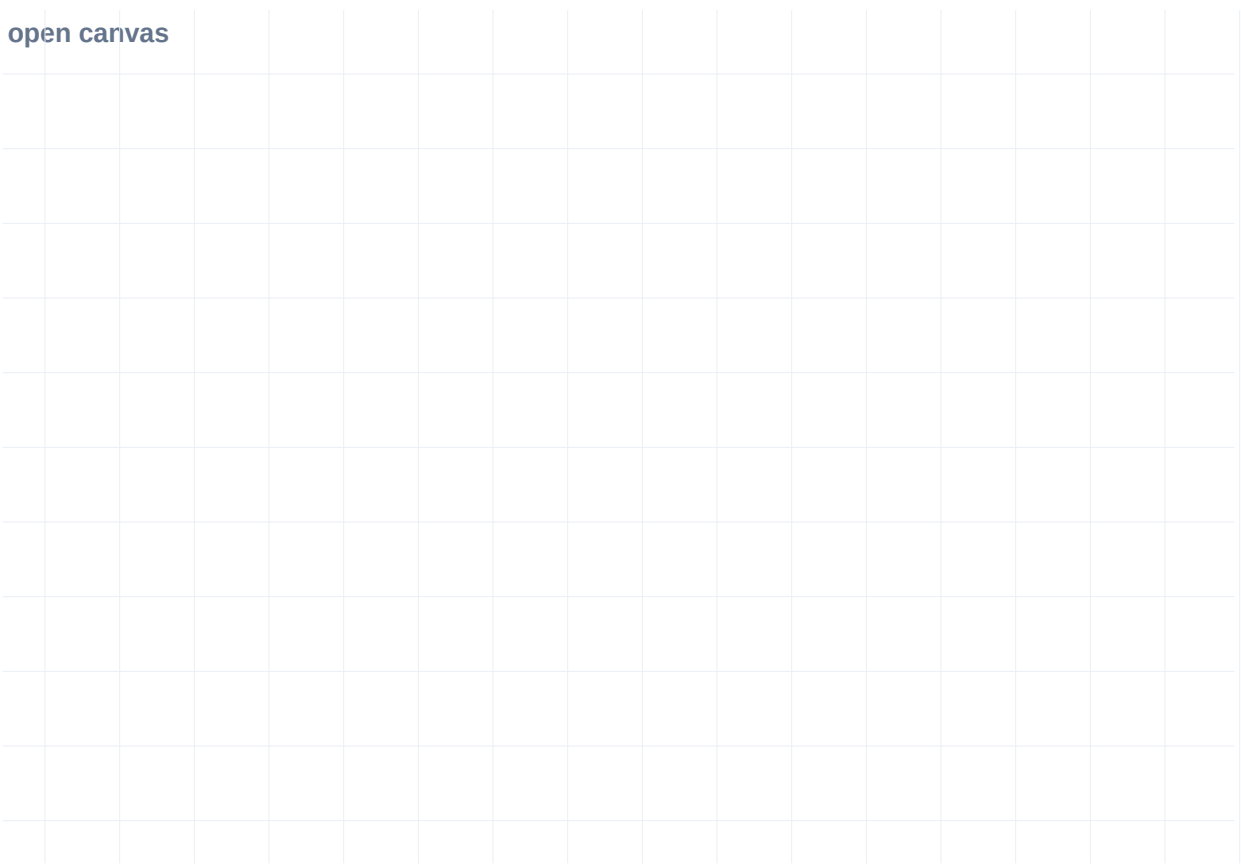
Logarithmic threshold

Show algebraic steps, graph verification, units, and interpretation.

EXPLAIN WHY

Do not only write the model.
Explain why this model fits
and where it fails.

open canvas

A large grid for graphing or drawing, consisting of 15 columns and 15 rows of squares. The grid is empty, with the text 'open canvas' in the top-left corner.

Rate comparison

Compare 3%, 5%, and 8% growth or decay. What happens to thresholds?

REFLECTION PROMPT

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

REVISION MARKER

Original idea:

Revised idea:

Reason for change:

Where does exponential behavior stop being realistic? What hidden

Do not only write the model.
Explain why this model fits
and where it fails.

Changing assumptions creates different mathematical realities. Compare at least three related model versions.

CLASS 4

Constraints, Capacity, and

How do physical and economic constraints reshape the system?

STRUCTURAL DESIGN LAB

Physical form creates algebraic structure.

Use this section as a field notebook. Let rough work, uncertainty, recalculation, and revision stay visible.

Flat blueprint

Original dimensions, square cutout variable, folding lines, units in centimeters.

REFLECTION PROMPT

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

REVISION MARKER

Original idea:

Revised idea:

Reason for change:

Height, interior dimensions, usable space, loading floor.

Do not only write the model.
Explain why this model fits
and where it fails.

Factored form, standard form, and why multiplying dimensions creates a cubic.

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Original idea:

Revised idea:

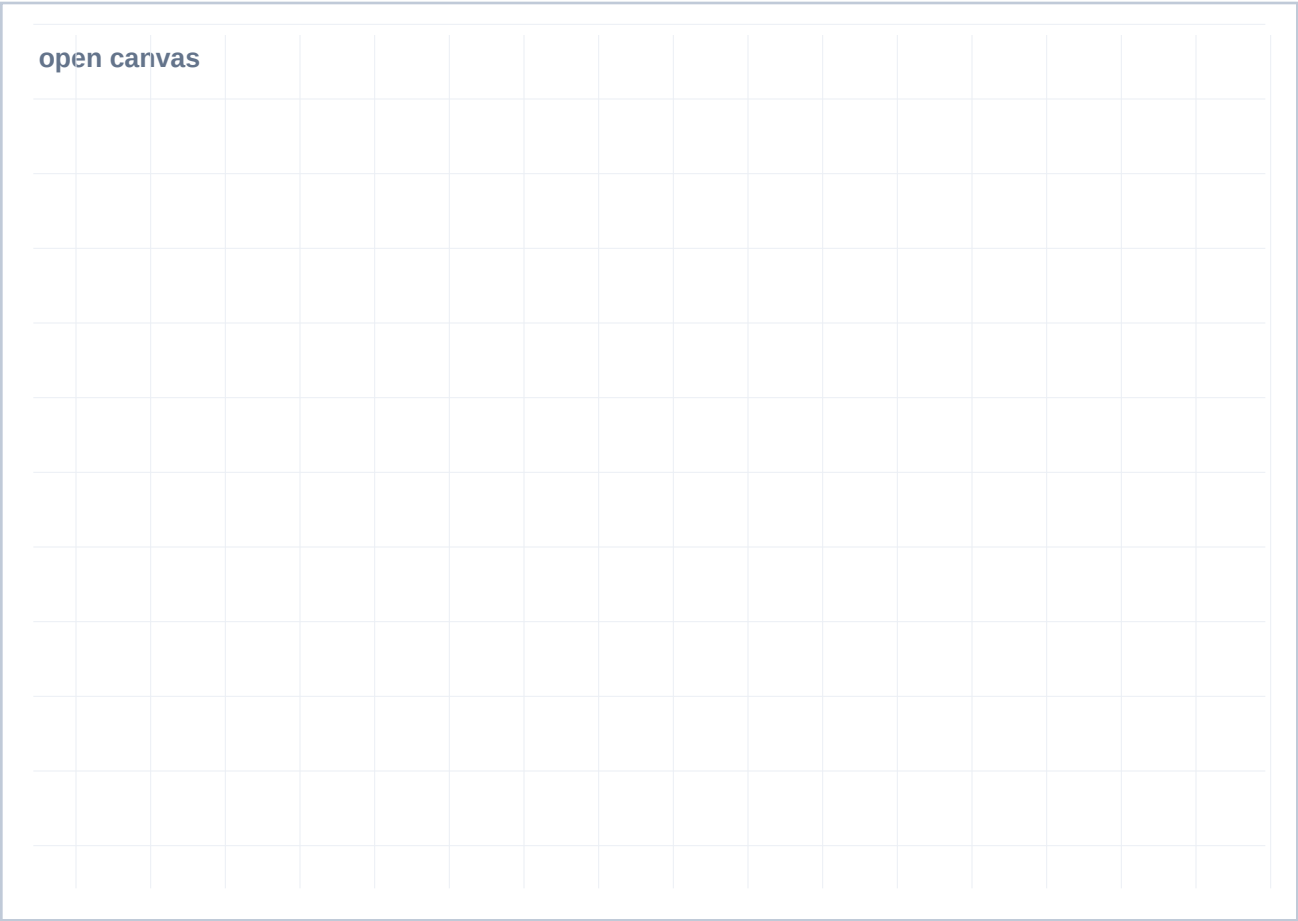
Reason for change:

Graph annotation

Intercepts, multiplicity, extrema, domain, unrealistic regions, physical meaning.

EXPLAIN WHY

Do not only write the model.
Explain why this model fits
and where it fails.



What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Reason for change:

Polynomial division

Dividend, divisor, quotient, remainder, and physical interpretation.

EXPLAIN WHY

Do not only write the model.
Explain why this model fits
and where it fails.

open canvas

Constraint revision

Change dimensions, cost, refrigeration, transport, or available space.
Compare results.

REFLECTION PROMPT

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

REVISION MARKER

Original idea:

Revised idea:

Reason for change:

Changing assumptions creates different mathematical realities. Compare at least three related model versions.

Polynomial interpretation studio

Use this page to connect polynomial structure to physical meaning.

Factored form

What does it reveal about dimensions, intercepts, or zero-volume situations?

Standard form

What does it make easier to analyze or compare?

Domain

Which x-values are mathematically valid but physically unrealistic?

end behavior, extrema, multiplicity, and physical interpretation

CLASS 5

Revision, Synthesis, and Conclusion

SYNTHESIS STUDIO

Evidence becomes a claim.

What story does our mathematics tell?

Use this section as a field notebook. Let rough work, uncertainty, recalculation, and revision stay visible.

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Reason for change:

Compare mathematical lenses and assumptions.

Do not only write the model.
Explain why this model fits
and where it fails.

Use evidence, assumptions, limitations, tradeoffs, and implications.

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Original idea:

Revised idea:

Reason for change:

Choose a format. Plan how mathematical evidence will be visible.

Do not only write the model.
Explain why this model fits
and where it fails.

What changed in your thinking? What assumption is doing the most work? What evidence would make this stronger?

Original idea:

Revised idea:

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